

SehaKotkuKaya

2025-11-30

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2     4.0.1      v tibble     3.3.0
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(scales)
```

```
##
```

```
## Attaching package: 'scales'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##     discard
```

```
##
```

```
## The following object is masked from 'package:readr':
```

```
##
```

```
##     col_factor
```

```
library(ggplot2)
```

```
library(stringr)
```

```
options(scipen = 999)
```

```
IMDB_Top_250_Movies <- read.csv("IMDB Top 250 Movies.csv", check.names = FALSE)
```

```
plot_data <- IMDB_Top_250_Movies %>%
```

```
  mutate(budget_raw = as.numeric(str_replace_all(budget, "[^0-9]", "")),
```

```
  budget_numeric = case_when(
```

```
    name == "Princess Mononoke" ~ 23500000,
```

```
    name == "3 Idiots" ~ 12000000,
```

```
    name == "Akira" ~ 10000000,
```

```
    name == "Spirited Away" ~ 19000000,
```

```
    name == "Dangal" ~ 9700000,
```

```
    name == "Grave of the Fireflies" ~ 3700000,
```

```

    TRUE ~ budget_raw
  ),

  box_office_clean = as.numeric(str_replace_all(box_office, "[^0-9]", ""))
) %>%

filter(!is.na(budget_numeric) & budget_numeric > 0)

ggplot(plot_data, aes(x = budget_numeric, y = rating)) +
  geom_point(alpha = 0.5, color = "darkred") +
  geom_smooth(method = "lm", col = "orange", size = 1.2) +

  scale_x_log10(
    breaks = c(1e6, 1e7, 1e8, 1e9),
    labels = c("1M", "10M", "100M", "1B")
  ) +

  labs(
    title = "Relationship Between Budget and IMDb Rating",
    x = "Budget (Log Scale, USD)",
    y = "IMDb Rating",
    caption = "Entries marked as 'Not Available' in the budget
              column (38 films) were filtered out prior to analysis."
  ) +
  theme_minimal()

```

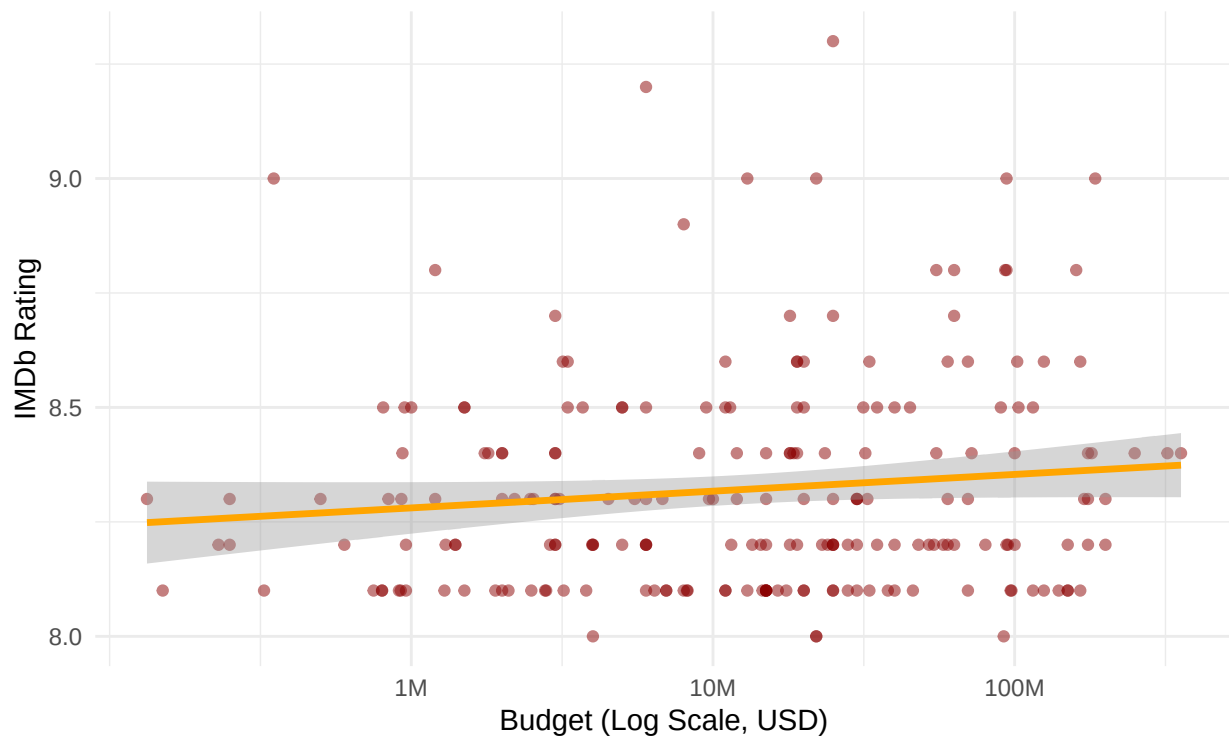
```

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## `geom_smooth()` using formula = 'y ~ x'

```

Relationship Between Budget and IMDb Rating

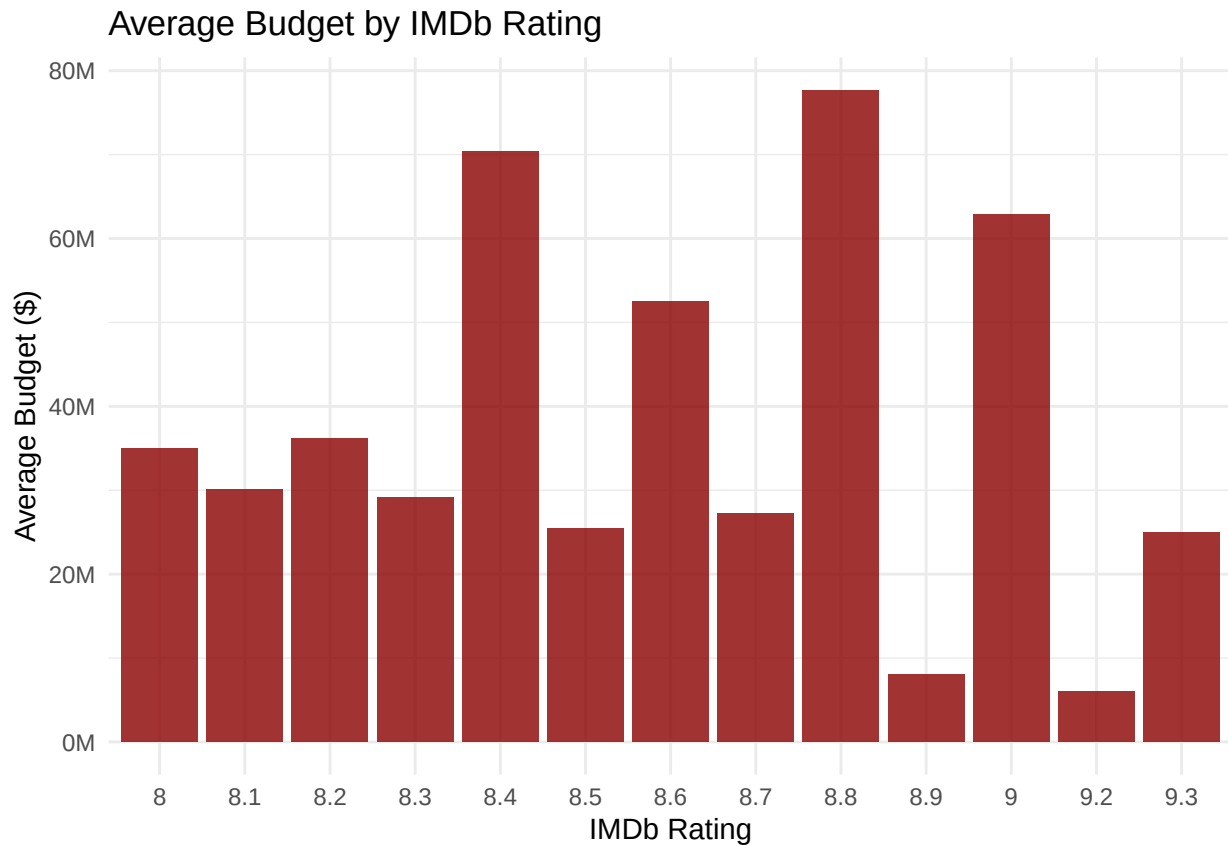


Entries marked as 'Not Available' in the budget column (38 films) were filtered out prior to analysis.

```
ggplot(plot_data, aes(x = factor(rating), y = budget_numeric)) +
  stat_summary(fun = "mean", geom = "bar", fill = "darkred", alpha = 0.8) +

  labs(
    title = "Average Budget by IMDb Rating",
    x = "IMDb Rating",
    y = "Average Budget ($)"
  ) +

  scale_y_continuous(labels = function(x) paste0(x / 1e6, "M")) +
  theme_minimal()
```



```
roi_data <- plot_data %>%
  filter(!is.na(box_office_clean) & box_office_clean > 0)

ggplot(roi_data, aes(x = budget_numeric, y = box_office_clean)) +

  geom_point(aes(color = rating), alpha = 0.7, size = 3) +

  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "black") +

  scale_x_log10(
    breaks = c(1e6, 1e7, 1e8, 1e9),
    labels = c("1M", "10M", "100M", "1B")
  ) +
  scale_y_log10(
    breaks = c(1e6, 1e7, 1e8, 1e9),
    labels = c("1M", "10M", "100M", "1B")
  ) +

  scale_color_gradient(low = "orange", high = "darkred") +

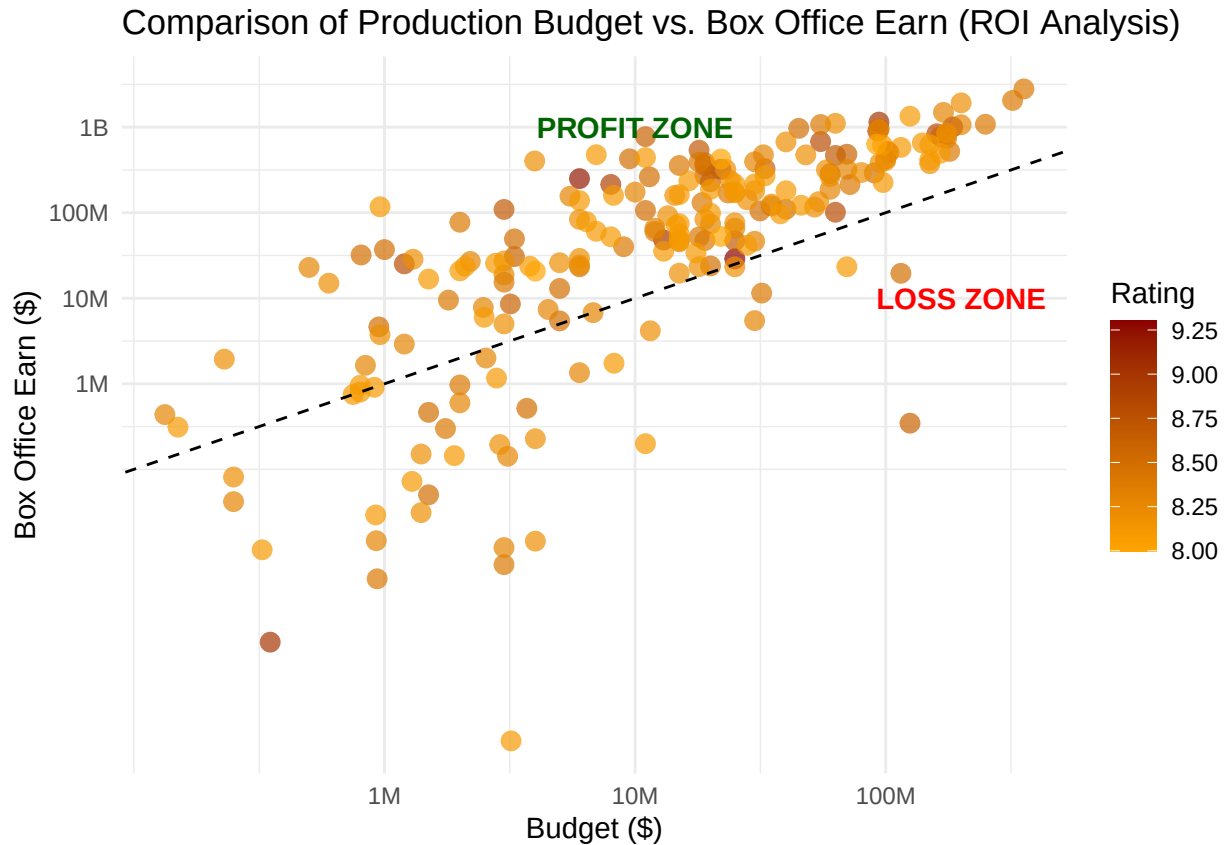
  labs(
    title = "Comparison of Production Budget vs. Box Office Earn (ROI Analysis)",
    x = "Budget ($)",
    y = "Box Office Earn ($)",
    color = "Rating"
  ) +
```

```

annotate("text", x = 1e7, y = 1e9, label = "PROFIT ZONE", color = "darkgreen", fontface = "bold") +
annotate("text", x = 2e8, y = 1e7, label = "LOSS ZONE", color = "red", fontface = "bold") +

theme_minimal()

```



- First one scatter graph show that there is no direct and strong correlation between the amount of money spent on a movie and it's IMDb rating (quality) so we cant say more money, higher the rate.
- The second chart shows that the best films in cinema history are not always the most expensive ones.
- The third chart (ROI plot) analyzes the commercial success of the movies. The fact that the 'Profit Zone' suggests that movies in the IMDb Top 250 list are generally successful not only artistically but also commercially.